

PROJECT PLANNING PHASE

Project Planning (Product Backlog, Sprint Planning, User Stories & Story Points)

Date	31 July 2026
Name	Kesireddy Padma
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	5 Marks

1. Product Backlog, Sprint Schedule, and Estimation

The product backlog below maps the implementation of the Metro Ticket Generating System in ServiceNow. The stories are prioritized according to the core booking workflow: station selection, ticket details, fare calculation, payment status, QR ticket generation, and validation.

Priority	User Story	Key Implementation	Story Points	Sprint	Status
P1	As passenger a, I want to select source and destination stations so that I can book a metro journey.	Service Catalog variables; station mapping	3	Sprint 1	Completed
P1	As a passenger, I want to enter the number and type of tickets so that the system can calculate the booking correctly.	Ticket count and passenger/ticket type variables	2	Sprint 1	Completed
P1	As a passenger, I want the fare to be calculated automatically based on the selected stations and journey details.	Client Script / fare logic	3	Sprint 1	Completed
P1	As a passenger, I want to make/select a payment status	Payment status field and booking flow	2	Sprint 2	Completed

	so that my ticket booking can be recorded.				
P1	As a passenger, I want a QR code generated for my confirmed ticket so that I can use it at the metro entry/exit.	QR generation and ticket confirmation	5	Sprint 2	Completed
P2	As a metro administrator, I want station details maintained in a dedicated table so that station information is reusable.	Metro Station Details table	3	Sprint 2	Completed
P2	As a passenger, I want the ticket to show booking date and expiry so that I know the validity period.	Booking date and ticket expiry fields	2	Sprint 2	Completed
P2	As a passenger, I want smart-card recharge details available so that recharge activity can be managed.	Smart-card recharge field/workflow	2	Sprint 2	Completed

Sprint Schedule

Sprint	Duration	Focus	Story Points	Expected Outcome
Sprint 1	10 days	Catalog form, station selection, ticket details, fare calculation	8	Working metro booking form with automatic fare
Sprint 2	10 days	Payment status, QR generation, station table, validity and recharge	14	Complete digital ticket generation workflow

2. Burndown Chart & Progress Metrics

A burndown chart can be used to track the remaining story points against the 10-day sprint cycle. For this project, progress is measured by completion of the ServiceNow configuration, scripting, ticket-generation, and validation tasks.

Sprint 1 Burndown Analysis:

- Starting Baseline: 8 Story Points.
- Mid-Sprint Status (Day 5): Completed the core Service Catalog form, station-selection variables, and ticket-detail configuration.
- Sprint Closure: Completed automatic fare calculation and the core booking workflow by the end of the sprint.

Sprint 2 Burndown Analysis:

- Starting Baseline: 14 Story Points.
- Mid-Sprint Status (Day 5): QR-code generation and payment-status validation required additional testing and debugging.
- Sprint Closure: Completed QR ticket generation, Metro Station Details mapping, booking date/expiry handling, and smart-card recharge functionality.

Progress Metrics

Metric	Sprint 1	Sprint 2	Overall
Planned Story Points	8	14	22
Completed Story Points	8	14	22
Outstanding Story Points	0	0	0
Completion	100%	100%	100%

3. Individual Sprint Retrospective

A self-retrospective was conducted after the planning and implementation iterations to evaluate development progress, identify technical challenges, and improve the next execution cycle.

What Went Well:

- Platform Efficiency: ServiceNow Service Catalog, tables, client scripts, and platform configuration reduced the need to build a web interface from scratch.
- Clear User Stories: Breaking the metro booking requirements into smaller stories made the implementation easier to track.
- Automation: Automatic fare calculation and QR-code ticket generation improved the digital booking workflow.
- Reusable Station Data: Maintaining station information in a dedicated Metro Station Details table supports consistent station mapping.

What Could Be Improved:

- Script Debugging: Client-side and QR-related scripts may require additional testing when field values or ServiceNow record identifiers do not match expected values.
- Data Validation: Station, ticket count, fare, payment status, and expiry values should be validated more thoroughly before ticket confirmation.
- Integration Planning: QR generation and payment-related functionality should receive additional time for testing in future estimates.

Action Items for the Next Phase:

1. Implement automated test scenarios for ticket booking, fare calculation, QR confirmation, and invalid station selections.
2. Add stronger validation for payment status, ticket expiry, and passenger input values.
3. Review and optimize client scripts and server-side logic for reliability and maintainability.
4. Create dashboards/reports to analyze bookings, fares, ticket types, and station usage.